Rikan Saha



Specialist, Genpact, Kolkata, India
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Carrier Objective

Results-driven Analytics & AI Engineer with 4 years of experience in Model Risk Management (MRM), with prior experience supporting North American banking clients and a current focus on AI-driven analytics solutions. Specializing in **Python-driven data analytics, Model Development, Power-BI & Excel-based regulatory reporting, and AI-enabled automation**. Designed and developed end-to-end analytics and AI solutions across risk and product domains, including automated pipelines, LLM and Agentic AI–based applications, and interactive dashboards for insight generation and workflow automation, with a keen interest in building scalable data-driven analytics solutions and intelligent decision-support systems.

Professional Experience

Responsibilities:

- **Specialist | *Genpact*** November 2024 - Present
  - Serving as a Subject Matter Expert (SME) in the Model Risk Management (MRM) domain for AI-driven product development initiatives within the Risk Analytics and Management service line, contributing to end-to-end implementation and deployment of AI/ML-enabled analytics solutions, including requirement analysis, feature design, and User Acceptance Testing (UAT).

- **Business Analyst | *Genpact*** September 2022 - October 2024
  - Led model risk aggregation, assessment, and regulatory reporting for a major North American bank, while managing model inventory systems and delivering automated, audit-ready reporting solutions. Developed Python- and Agentic-AI-driven solutions, including dashboards, data pipelines, and AI applications, to enable insight generation, workflow automation, and data-driven decision-making.
- **Awards and Achievements**: Received two performance recognitions for automating key operational processes (Python, MS-Excel, Power-BI) and delivering high-quality model inventory enhancements while efficiently managing quarter-end volume spikes in regulatory reporting cycle.

Projects

- **Agentic AI Review Attestation Assistant |** *Python, SQLite, LangGraph + RAG, Streamlit, Azure OpenAI*
  - Designed and developed a web-based application to streamline the annual model review attestation process using Python, SQLite, Streamlit, and an agentic LLM + RAG-based pipeline, enabling end-to-end workflow execution through an interactive frontend while utilizing optimized SQL queries and efficient backend processing for scalable automation and context-aware recommendations.
  - Enabled efficient information retrieval and workflow automation through an integrated chatbot interface querying internal and external data sources, improving data access, information consistency across Model Developer and Validator reviews, and overall process efficiency.

- **Enterprise Product Intelligence Platform |** *Python, FastAPI, React, NLP, LangGraph + RAG, Azure OpenAI*
  - Designed and deployed an AI-powered Product Intelligence Platform to automate large-scale customer feedback analysis by implementing an end-to-end AI pipeline with semantic embeddings, K-Means clustering, LLM-based summarization, and RAG, enabling scalable extraction of customer pain points and product insights.
  - Implemented a LangGraph-based multi-agent architecture integrating Recommendation, Business Impact, and Validation agents with a FAISS-backed knowledge base to generate grounded, explainable, and business-prioritized recommendations, reducing manual review effort and accelerating data-driven product decisions through a FastAPI and React application.

- **AI-Enabled Model Risk Analytics Platform |** *Python, SQLite, LLM, Power-BI*
  - Designed and developed an automated model lifecycle reporting and analytics solution to support Model Risk Management (MRM) governance workflows (KRIs, MRMC, MSMM), using object-oriented Python, SQLite, Power-BI, and LLM-based techniques, incorporating multi-source data integration and data quality enhancement processes.
  - Enabled automated reporting and agent-based cross-report analysis, using LLM-powered reasoning, predefined business rules, key risk drivers, and contributing factors to generate risk insights aligned with Model Risk Appetite thresholds and regulatory reporting requirements.

- **Credit Risk Modeling & Expected Credit Loss (ECL) Framework |** *PySpark (pyspark.ml), Plotly, Ml-Flow*
  - Designed and developed an end-to-end Credit Risk Modeling solution to estimate Probability of Default (PD), Loss Given Default (LGD), Exposure at Default (EAD), and Expected Credit Loss (ECL) using PySpark-based distributed data processing, feature engineering, and Logistic Regression modeling on large-scale financial datasets.

- Enabled automated risk scoring, borrower segmentation, expected loss estimation, and portfolio risk analytics by integrating PD-LGD-EAD frameworks with interactive Plotly-based visualizations for risk storytelling, exposure analysis, and model performance interpretation.

- **Market Regime Analytics & Forecasting System** | *Python (OOP), SQLite, Scikit-Learn, Statsmodels, Power-BI*
  - Designed and developed an end-to-end macroeconomic analytics solution to identify market regimes (Expansion, Recession, Stagflation) using clustering on IMF data, incorporating feature engineering, K-Means–based regime detection, and time-series forecasting for GDP growth, inflation, and unemployment.
  - Enabled forward-looking economic insights and trend analysis by integrating outputs into a SQLite database and delivering interactive Power-BI dashboards for regime visualization and decision support.

- **Additional AI Engineering Projects:** Developed additional AI-driven applications, including an Agentic Multi-Index RAG System, an AI-Powered Data Quality Accelerator, and an AI-Based Coding Interface, leveraging FastAPI, React, LangGraph, RAG, Azure OpenAI, and FAISS to build intelligent document retrieval systems, metadata-driven indexing, AI-assisted code generation, workflow automation, scalable application architectures, and developer productivity solutions.

Research Interest and Publications

Algorithm Design and Analysis, Machine Learning, Statistical Analysis, Reinforcement Learning.
Publication: S. Roy, **R. Saha**, S. Sarkar, R. Mehera, R. K. Pal and S. K. Bandyopadhyay, "Brain Tumour Segmentation Using S-Net and SA-Net," in IEEE Access, vol. 11, pp. 28658-28679, 2023, doi: *10.1109/ACCESS.2023.3257722*.

Proficiencies

- **Core Analytical & AI Concepts:** Model Development, Risk Analytics, Data Analysis & Interpretation, Statistical Analysis, Exploratory Data Analysis (EDA), Application Programming Interface (REST API), Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing (NLP), Time Series Forecasting, Generative AI, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic AI, Vector Databases (FAISS, pgvector).
- **Programming Languages & Tools:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn, Bokeh, Transformers, LangChain, LangGraph, FAISS, Openpyxl, python-docx, Streamlit), Open API, JavaScript, SQL, Power-BI, MS Excel.
- **Regulatory Frameworks:** Supervision and Regulation Letter (versions SR 11-7, SR 26-2).
- **Development Environments & Platforms:** Git, GitHub, Microsoft Azure (Azure AI Foundry, Azure OpenAI, Azure App Service, Azure Static Web Apps, Azure Container Apps), Hugging-Face, GitHub Copilot, Claude, Microsoft Copilot, OpenAI, Visual Studio Code, Anaconda, Docker.
- **Core Computer Science Fundamentals:** Data Structures & Algorithms (DSA), Database Management Systems (DBMS), Object-Oriented Programming (OOP), Unified Modeling Language (UML).
- **Additional Tools & Programming Languages:** HTML, CSS, Java, C, C++, LaTeX, Microsoft Office Suite.

Academic Qualifications

- **M.Tech. in Computer Science & Engineering** | *University of Calcutta (UCSTA)* 2020 - 2022
- **M.Sc. in Computer Science** | *University of Calcutta (UCSTA)* 2018 - 2020
- **B.Sc. in Computer Science** | *West Bengal State University* 2015 - 2018

Academic Research Projects

- **M.Tech. Project** | Image Semantic Segmentation using Reinforcement Learning 2022
- **M.Sc. Project** | Target Coverage & Connectivity using ILP and Graph Algorithm 2020

Online Certifications

- Advanced LangChain Techniques: Mastering RAG Application, Udemy. LINK
- LangChain- Develop AI Agents with LangChain & LangGraph, Udemy. LINK
- The Agentic AI Engineering Masterclass 2026, Udemy. LINK
- Vector Databases Fundamentals, Udemy. LINK
- LLM Fine Tuning on OpenAI, Udemy. LINK
- Hands-on approach to AI for real-world applications, IIT Kharagpur AI4ICPS Certificate Programme. LINK
- Credit Risk Modeling in Python, Udemy. LINK
- Mathematics - Basic to Advanced for Data Science and Gen AI, Udemy. LINK
- Natural Language Processing with Python, Udemy. LINK
- JavaScript Basic for Beginners, Udemy. LINK